



Shining a Light on the Extensive Nightly Activities of *Callistoctopus furvus* in South Caicos

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Abstract: *Callistoctopus furvus* is a nocturnal species of octopus that inhabits the shallow water marine ecosystems of the Western Atlantic. *C. furvus* is a mesopredator, allowing it to act as a bridge between trophic levels. The rapidly changing skin pattern and behavior of *C. furvus* facilitates its survival through crypsis, which is the ability of an organism to avoid detection through mechanisms like camouflage and mimicry. Understandings of skin patterning and behavior is hindered as previous observations have been made with white light, which is known to disrupt behavior in nocturnal animals. We explored the relationships between skin patterning, substrate, and behavior using both white and red light. We found that certain skin patterns were associated with particular substrates and behaviors, likely in order to evade discovery through crypsis or intimidate other species through deimatic behaviors. We also found that *C. furvus* tended to be more stationary under red light, suggesting that this light color was less disruptive. Greater knowledge of skin patterning, behaviors and their correlations to movement, helps in interpreting interactions with other organisms and promoting future behavioral research.

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1. Introduction

Octopuses are an essential species in marine ecosystems, acting as a trophic bridge as they prey on benthic fish and are preyed upon by larger species inhabiting the pelagic zone (Hair, 2023; Borges et al., 2023). Without octopuses, lower trophic population levels may increase, and higher trophic levels populations decline, alarmingly affecting the whole ecosystem. In the face of increasing anthropogenic pressures on marine ecosystems, it is crucial to develop knowledge of octopuses' behaviors in order to determine their full impact on the ecosystems they inhabit.

Octopuses generally live solitary, nocturnal, and nomadic lives on the seafloor which serves as a challenge when observing their natural behavior (Mather & Anderson, 1993). External light sources are required to perform such observations but artificial light heavily impacts behavior of benthic organisms, as it tends to disrupt natural behavior (Geoffroy et al., 2021). Artificial light can potentially blind organisms, preventing them from exhibiting appropriate defense and escape mechanisms. By contrast, red light is less disruptive to nocturnal animal behavior as it is not perceived by most organisms (Weiss et al., 2006).

Despite the difficulties that come with studying nocturnal octopuses, shallow-water octopuses are known to be major predators of numerous species, using multiple foraging techniques to capture their prey. However, because octopuses generally spend the majority of their time in their dens (e.g. Mather, 1988; O'Brien et al., 2023), little is known about individual hunting tactics and their overall feeding and foraging strategies. Octopuses utilize saltatory (stop-and-go) hunting as a method to help locate and catch their prey. Typically, in saltatory foraging, the species will stay in a stationary position to scan a space for prey and only use moments between pauses to move into an unscanned space. Using this strategy helps to maximize search time and avoid predation. The risk of predation plays a significant role in this foraging strategy, and crypsis through skin patterning is vital to minimizing predation risk while optimizing prey capture (Leite, Haimovici, & Mather, 2009).

The combination of chromatophores (pigment cells), iridophores (reflective cells) and papillae (patches of skin that can be raised or lowered) allow cephalopods to rapidly change their appearance (Hanlon and Messenger, 2018) in order to blend in with their surrounding environment, break up the outline of their body, and mimic other objects (Hanlon and Messenger, 2018; Josef et al., 2012). Changing skin patterns can also act as communication signals; for example, skin patterns can be used to threaten other species or attract mates (How et al., 2017). The evolution of rapidly changing skin patterns in octopuses are imperative towards their survival and interactions with other organisms.

Callistoctopus furvus is a species of nocturnal octopus present in the Western Atlantic that was recently differentiated from *C. macropus* in the Eastern Atlantic (Jesus et al., 2021). *C. furvus* live in shallow waters with temperatures ranging from 23°C to 29°C. They are often found in waters less than five meters deep, and typically inhabit sandy plains, seagrass beds and coral reefs (Jesus et al., 2021; Jesus et al., 2022). Due to its recent discovery and solitary behavior, little is known about this species, necessitating research.

We conducted a field and video-based study with three main questions: 1) How do various substrates relate to the skin patterning of *C. furvus*? 2) How do different behaviors relate to the skin patterning of *C. furvus*? 3) How will *C. furvus* react to being exposed to red versus white light? We anticipate *C. furvus* will appear lighter (i.e. Mottle or Stippled) on lighter substrates (i.e. Sand, Bedrock or Boulder, Loose Rock) and more Brick-Red in the presence of darker substrates (i.e. Coral, Red Algae) to better camouflage with its surroundings. We also predict *C. furvus* will present a mixed color appearance (i.e. Brick Red and White Elements, Mottle, Racing Stripe, Stippled) when Stationary, Crawling, going through Bipedal Locomotion, Entering a Hole, or Burrowing to better prevent detection by other organisms as the octopus will be able to blend into a higher variety of substrates. When foraging for food or escaping from predators (i.e. Arm Exploration, Jetting, Parachute Attack), *C. furvus* will appear Brick Red with White Arm Spots or Polka Dot as a type of deimatic behavior, mimicking more dangerous species and intimidating its predators (Hanlon and Messenger, 2018). We hypothesize that *C. furvus* will demonstrate more defense-driven behaviors (eg. Burrowing, Entering Hole, Jetting) when being filmed

with white light and more natural behavior (eg. Arm Exploration, Crawling, Stationary) in the presence of red light due to the disturbance each light color causes. The results of our study provide information about behaviors and skin patterning which can be used in future research of *C. furvus* to help understand their interactions with other species and role in the environment.

2. Methods

2.1 Sites

Survey sites were located on South Caicos, an island in the Caribbean (21.5112° N, 71.5190° W). The predominant substrates found in South Caicos' waters include but are not limited to coral reefs, seagrass beds, and sandy plains. Site depths range from one to eight meters and sites were chosen based on previous sightings of *C. furvus*. A total of 9 sites were visited in November 2023. Previous data from July 2020 to April 2023 were also analyzed (Figure 1).

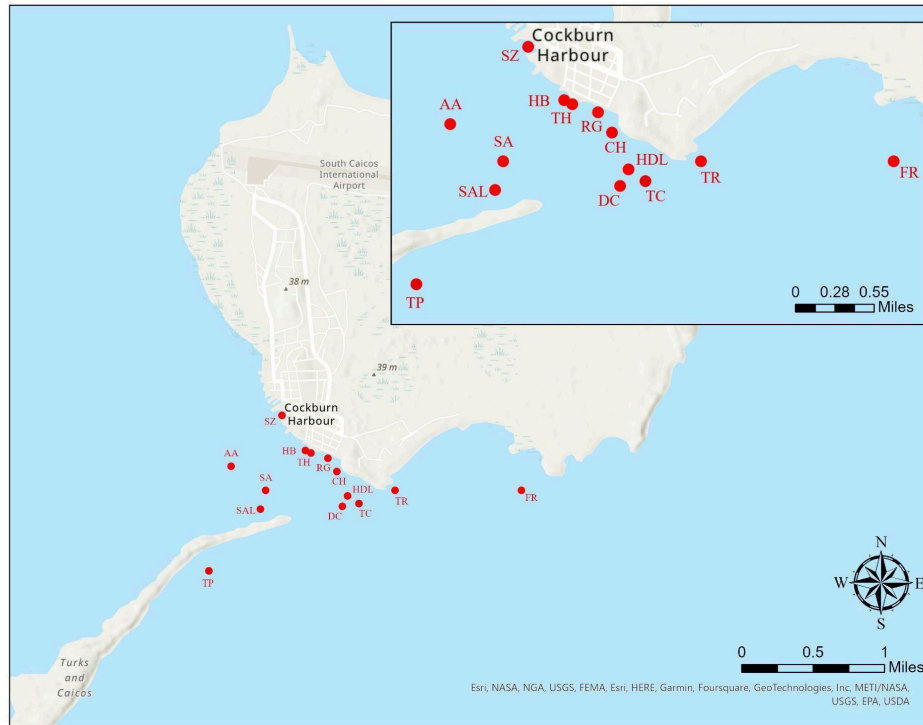


Figure 1. Map of 14 research sites from July 2020 to November 2023. SZ = Swim Zone, AA = Admiral's Aquarium, SA = Shark Alley, SAL = Shark Alley Lagoon, HB = Hotel Beach, TH = Turtle House, RG = Regatta, CH = Cox Hotel, HDL = Huey Dewey Louie, TC = The Chain, DC = Dove Cay, TP = The Plane, TR = Tucker's Reef, FR = Fish Rock.

2.2 Field Data Collection

Videos were collected while snorkeling and SCUBA diving using Nikon CoolPix W300 underwater cameras with 5000K SeaDragon lights. While diving at night (approximately 19:30 - 21:00) researchers searched for *C. furvus* in a diamond formation with a dive leader navigating, followed by three to six research divers in a horizontal line, followed by a safety diver (Figure 2). All researchers scanned and searched with white light until *C. furvus* was found, then researchers switched to red light.

Once filming began, white lights were occluded, depth and time were noted, and the remaining buddy teams moved away from *C. furvus* with the safety diver to ensure minimal disturbance to the octopus. *C. furvus* were filmed until they were no longer able to be tracked by the filmer (i.e. octopus jetted away or entered a hole) or until any diver reached 1700 PSI. Dives ranged from 40 to 70 minutes.

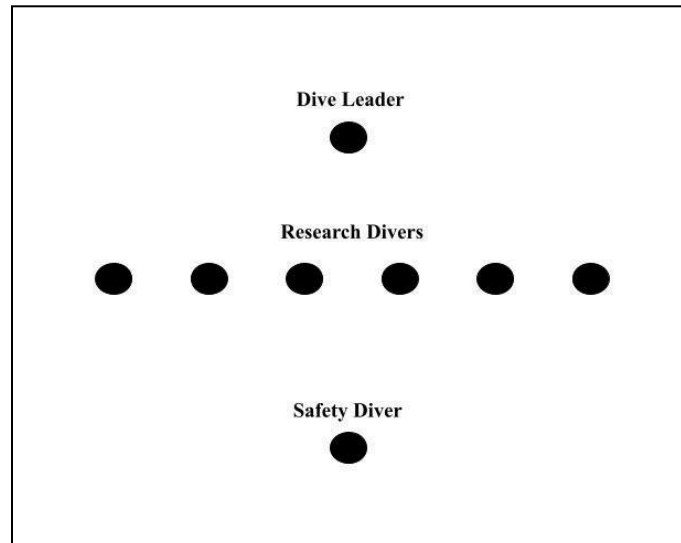


Figure 2. Dive formation diagram.

2.3 Video Data Collection

White and red light videos were watched. For white light videos, second-by-second analyses of behavior, skin pattern, and substrate types were documented. For red light videos, only behavior and substrate types were reported. Behaviors were categorized as: Stationary, Crawling, Jetting, Parachute Attack, Bipedal Locomotion, Burrowing, Entering Hole, Arm Capture, and Arm Exploration (see descriptions in Table 1). According to Figure 3, skin patterns were classified as: Stippled, Mottle, Brick Red with White Arm Spots, Brick Red and White Elements, Racing Stripe, and Polka Dots. Substrate types included: Loose Rock, Bedrock or Boulder, Sand, Live Coral, Dead Coral, Sponge, Sparse Seagrass, Seagrass, Dead Seagrass, Green Algae, Brown Algae, Red Algae, and Knocked Conch Shell. All videos were analyzed by two individuals; analyses were then compared and videos were rewatched to rectify differences.

Table 1. Behavior ethogram of *C. furvus* (O'Brien and Lekas, in preparation).

Behavior	Description
Stationary	Octopus remains in one location, with only motion being ventilation or small arm movements
Crawling	Octopus uses arms to move across the substrate
Jetting	Octopus shoots water from siphon to rapidly move away
Parachute Attack	Octopus engulfs a rock or coral head with its arms and web
Bipedal Locomotion	Octopus moves across substrate on two back arms, usually while raising front two arms up in a corkscrew shape
Burrowing	Octopus uses funnel and/or arms to bury itself in the sand
Entering Hole	Octopus squeezes into an existing hole in the substrate
Arm Capture	Octopus captures prey with one or more arms
Arm Exploration	Octopus uses arms to search crevices, with the rest of the body mostly stationary

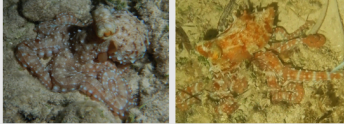
Skin Pattern	Photograph
1.) Stippled	
2.) Mottle	
3.) Brick Red with White Arm Spots	
4.) Brick Red and White Elements	
5.) Racing Stripe	
6.) Polka Dots	

Figure 3. The six major skin patterns of *C. furvus* (O'Brien and Lekas, in preparation).

2.4 Data Analysis

Data was analyzed using R (version 4.2.2). Chi-square goodness of fit tests were performed for all behavioral and substrate categories. All behavioral and substrate categories were significant, so the EMT package was used to perform post-hoc binomial tests for each skin pattern under each behavior and substrate type. P-values from binomial tests were adjusted to correct for type one error with the p-adjust Holm method. For comparison of behavior in white versus red light, data was tested for normality. If the behavior met parametric assumptions, a Welch two sample t-test was conducted. If the behavior did not meet parametric assumptions, or a significant difference was found, a Wilcoxon test was performed. The median (IQR) was also found for all behaviors except for Stationary and Unknown. All stacked bar charts were made using ggplot2.

3. Results

3.1 The Effect of Substrate on Skin Pattern

For substrate on skin pattern, all of the chi-square comparisons were significant: Bedrock or Boulder ($P < 0.0001$, X-squared = 305), Brown Algae ($P < 0.0001$, X-squared = 113), Green Algae ($P < 0.0001$, X-squared = 113), Loose Rock ($P < 0.0001$, X-squared = 102), Red Algae ($P < 0.01$, X-squared = 17.4), Sand ($P < 0.0001$, X-squared = 79.2), Sparse Seagrass ($P < 0.0001$, X-squared = 179), and Unknown ($P < 0.0001$, X-squared = 798). On Bedrock or Boulder, the skin pattern Brick Red with White Arm Spots (25.0%; CI 95%: 0.221-0.289; $N = 658$; binom.test, $P < 0.0001$) and Stippled (35.0%; CI 95%: 0.316 - 0.390; $N = 658$; binom.test, $P < 0.0001$) was significantly higher than the expected count of any other skin pattern. On Brown Algae, the observed skin pattern Brick Red and White Elements (32.0%; CI 95%: 0.258 - 0.380; $N = 237$; binom.test, $P < 0.0001$) and Mottle (46.0%; CI 95%: 0.391 - 0.521; $N = 237$; binom.test, $P < 0.0001$) were significantly higher than the expected counts of the other skin patterns. Similarly, on the Green Algae, the observed count for skin pattern Mottle (53.0%; CI 95%: 0.446 - 0.615; $N = 143$; binom.test, $P < 0.0001$) and Polka Dot (20.0%; CI 95%: 0.134 - 0.270; $N = 143$; binom.test, $P < 0.010$) were significantly higher than the expected count. The observed count on substrate Loose Rock were significantly higher for skin pattern Polka Dot (33.0%; CI 95%: 0.227 - 0.454; $N = 73$; binom.test, $P < 0.0001$), Racing Stripe (26.0%; CI 95%: 0.167 - 0.381; $N = 73$; binom.test, $P < 0.0001$), and Unknown (14.0%; CI 95%: 0.069 - 0.241; $N = 73$; binom.test, $P < 0.015$) than the expected count. For Red Algae, all of the skin patterns were lower than the observed count and not significant, except Mottle (100%, CI 95%: 0.590 - 1.00; $N = 7$; binom.test, $P < 0.010$). For substrate Sand, the observed count were significantly higher for skin pattern Mottle (35.0%; CI 95%: 0.301 - 0.422; $N = 256$; binom.test, $P < 0.038$), Polka Dot (18.0%; CI 95%: 0.135 - 0.232; $N = 256$; binom.test, $P < 0.010$), and Racing Stripe (13.0%; CI 95 %: 0.090 - 0.176; $N = 256$; binom.test, $P < 0.020$). Substrate Sparse Seagrass had a significantly higher observed count for skin pattern Brick Red and White Elements (40.0%; CI 95%: 0.338 - 0.462; $N = 253$; binom.test, $P < 0.0001$) and Racing Stripe (18.0%; CI 95%: 0.136 - 0.235; $N = 253$; binom.test, $P < 0.0001$). For the Unknown substrate, the observed count was significantly high

for the Unknown (100%; CI 95%: 0.00 - 0.074; N = 48; binom.test, $P < 0.0001$) skin patterns (Figure 4.)

The p-values for all substrate and skin patterns are listed in Table 2.

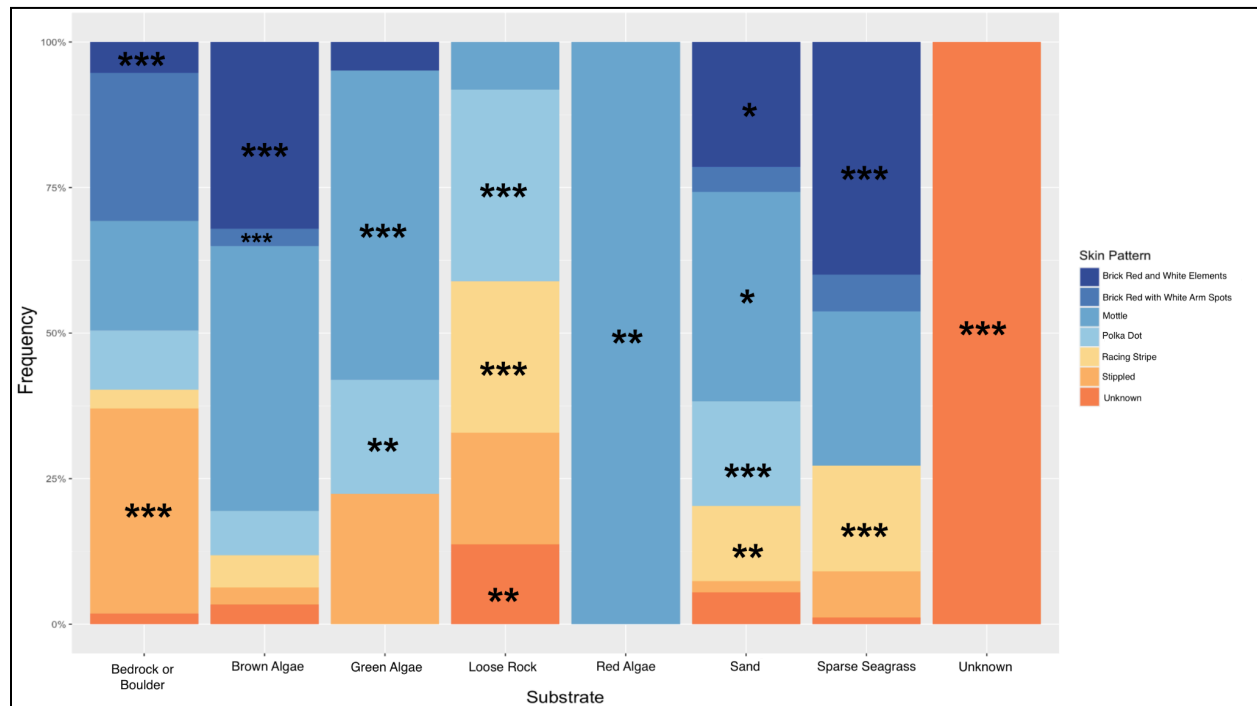


Figure 4. Stacked bar plot depicting the relative percentages of skin pattern on each substrate. Each color represents a skin pattern (Brick Red and White Elements, Brick Red with White Arm Spots, Mottle, Polka Dot, Racing Stripe, Stippled, and Unknown). Asterisks (*) represent counts significantly higher than the expected counts. One asterisk (*) represents $P < 0.05$, two asterisks (**) represents $P < 0.01$, and three asterisks (***) represents $P < 0.001$. No asterisks indicate no significantly higher difference between observed and expected counts.

Table 2. The table shows the adjusted p-values between observed and expected counts. The bolded values represent skin patterns with observed counts significantly higher than expected counts. The italicized values represent skin patterns with observed counts significantly lower than expected counts. Normally formatted values represent skin patterns with no significant difference between observed and expected counts.

	Bedrock or boulder	Brown algae	Green algae	Loose rock	Red algae	Sand	Sparse seagrass	Unknown
Brick red and white elements	<0.0001	<0.0001	<0.001	<0.0001	1.00	0.068	<0.0001	<0.010
Brick red with white arm spots	<0.0001	<0.0001	<0.0001	<0.0001	1.00	<0.0001	<0.010	<0.010
Mottle	<0.0001	<0.0001	<0.0001	<0.0001	<0.010	0.038	0.049	<0.0001
Polka dot	0.574	0.351	<0.010	<0.0001	1.00	<0.010	<0.0001	0.017
Racing stripe	<0.0001	0.351	<0.0001	<0.0001	1.00	0.020	<0.0001	0.031
Stippled	<0.0001	<0.0001	0.236	0.879	1.00	<0.0001	<0.0001	<0.001
Unknown	<0.0001	0.351	<0.010	0.015	1.00	1.00	<0.010	<0.0001

3.2 The Relationship Between Behavior on Skin Pattern

All chi-square comparisons for behavior by skin pattern were significant: Arm Exploration ($P < 0.0001$, X-squared = 39.7), Bipedal Locomotion ($P < 0.0001$, X-squared = 135), Burrowing ($P < 0.0001$, X-squared = 113), Crawling ($P < 0.0001$, X-squared = 90.2), Entering Hole ($P < 0.0001$, X-squared =

66.1), Jetting ($P < 0.0001$, X-squared = 336), Parachute Attack ($P < 0.0001$, X-squared = 30.9), Stationary ($P < 0.0001$, X-squared = 79.5), and Unknown ($P < 0.0001$, X-squared = 1350). The observed counts for the skin patterns Brick Red with White Arm Spots (25.0%; CI 95%: 0.155 - 0.375; N = 67; binom.test, $P < 0.012$) and Racing Stripe (19.0%; CI 95%: 0.108 - 0.309; N = 67; binom.test, $P < 0.012$) were significantly higher than the expected counts during Arm Exploration. The observed counts for the skin patterns Brick Red and White Elements (43.0%; CI 95%: 0.354 - 0.510; N = 165; binom.test, $P < 0.0001$) and Racing Stripe (13.0%; CI 95%: 0.086 - 0.195; N = 165; binom.test, $P < 0.041$) were significantly higher than the expected counts during Bipedal Locomotion. The observed counts for the skin pattern Mottle (100%; CI 95%: 0.920 - 1.00; N = 44; binom.test, $P < 0.0001$) were significantly higher than the expected counts during Burrowing. The observed counts for the skin patterns Racing Stripe (12.0%; CI 95%: 0.093 - 0.150; N = 530; binom.test, $P < 0.010$) and Stippled (31.0%; CI 95%: 0.270 - 0.351; N = 530; binom.test, $P < 0.0001$) were significantly higher than the expected counts during Crawling. The observed counts for the skin pattern Brick Red with White Arm Spots (34.0%; CI 95%: 0.248 - 0.434; N = 107; binom.test, $P < 0.0001$) were significantly higher than the expected counts while Entering Holes. The observed counts for the skin patterns Brick Red and White Elements (27.0%; CI 95%: 0.205 - 0.332; N = 200; binom.test, $P < 0.001$) and Polka Dot (49.0%; CI 95%: 0.414 - 0.557; N = 200; binom.test, $P < 0.0001$) were significantly higher than the expected counts while Jetting. The observed counts for the skin pattern Mottle (100%; CI 95%: 0.735 - 1.00; N = 12; binom.test, $P < 0.0001$) was significantly higher than expected counts during Parachute Attack. The observed counts for the skin patterns Brick Red with White Arm Spots (16.0%; CI 95%: 0.132 - 0.202; N = 462; binom.test, $P < 0.010$), Mottle (35.0%; CI 95%: 0.309 - 0.398; N = 462; binom.test, $P < 0.010$), and Stippled (26.0%; CI 95%: 0.222 - 0.305; N = 462; binom.test, $P < 0.001$) were significantly higher than the expected counts while Stationary. The observed counts for Unknown skin patterns (99.0%; CI 95%: 0.00 - 0.041; N = 88; binom.test, $P < 0.010$) were significantly higher than the expected count during Unknown behaviors (Figure 5). All other p-values are listed in Table 3.

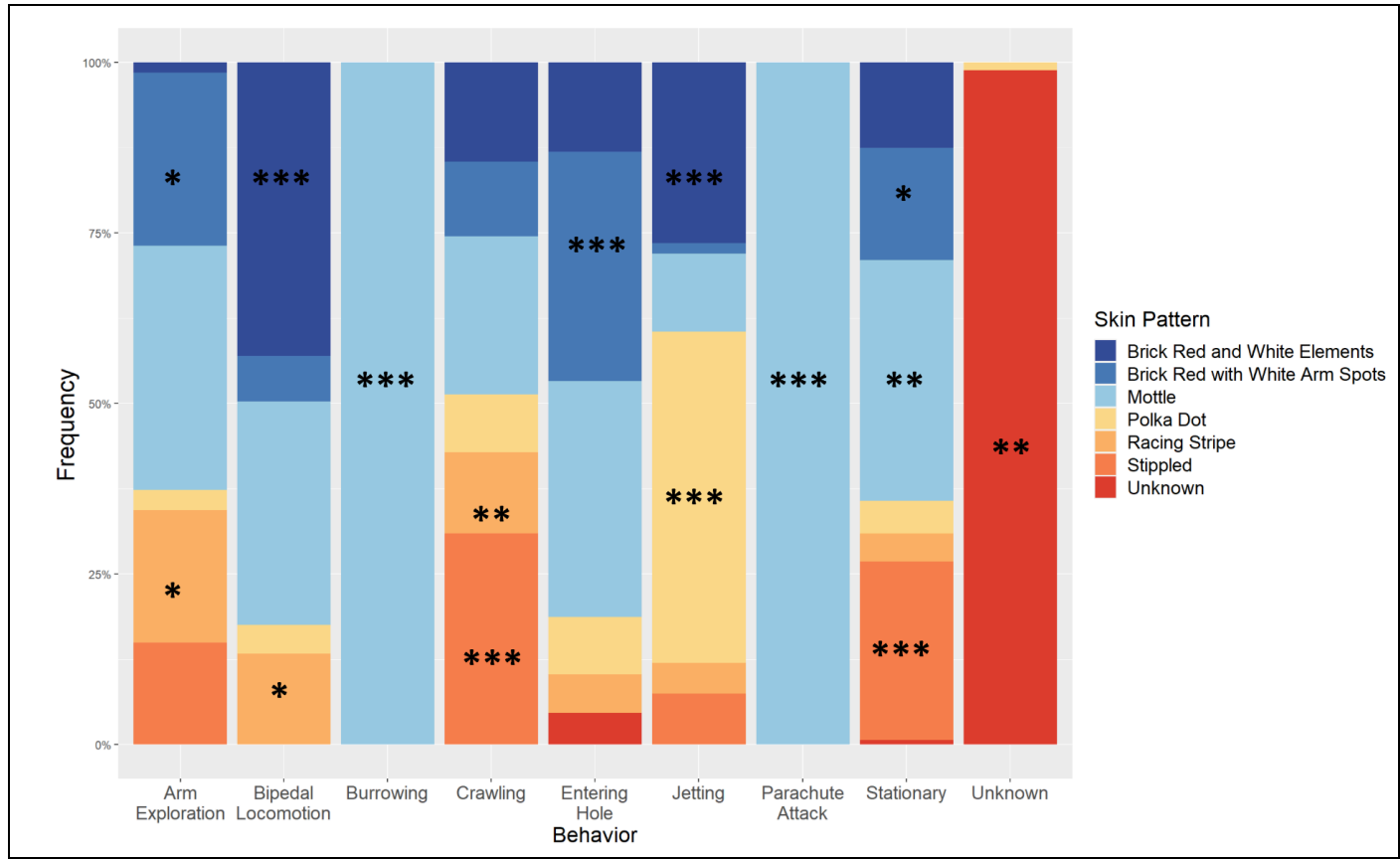


Figure 5. Stacked bar plot showing the relative frequency of skin pattern for each behavior. Asterisks represent counts significantly higher than the expected counts. One asterisk (*) represents $P < 0.050$, two asterisks (**) represents $P < 0.010$, and three asterisks (***) represents $P < 0.001$. No asterisks indicate no significantly higher difference between observed and expected counts.

Table 3. Table showing adjusted p-values of significant differences between observed and expected. Bolded values represent skin patterns with observed counts significantly higher than expected counts. Italicized values represent skin patterns with observed counts significantly lower than expected counts. Normally formatted values represent skin patterns with no significant difference between observed and expected counts.

	Arm Exploration	Bipedal Locomotion	Burrowing	Crawling	Entering Hole	Jetting	Parachute Attack	Stationary	Unknown
Brick Red and White Elements	<0.010	<0.0001	<0.010	0.530	1.00	<0.001	1.00	0.028	<0.0001
Brick Red with White Arm Spots	0.012	0.062	0.033	0.530	<0.0001	<0.0001	1.00	0.010	<0.0001
Mottle	0.447	0.263	<0.0001	0.021	0.995	<0.0001	<0.0001	<0.010	<0.0001
Polka Dot	0.123	0.015	0.038	0.244	1.00	<0.0001	1.00	<0.0001	<0.010
Racing Stripe	0.012	0.041	0.095	<0.010	1.00	0.087	1.00	<0.010	<0.010
Stippled	0.531	<0.0001	<0.010	<0.0001	<0.0001	<0.0001	0.841	<0.001	<0.0001
Unknown	0.123	<0.001	0.179	<0.0001	1.00	<0.0001	1.00	<0.0001	<0.010

3.3 The Effect of White versus Red Light on Behavior

Arm Exploration, Crawling, and Unknown behaviors did not meet parametric assumptions while Bipedal Locomotion, Burrowing, Entering Hole, Jetting, Parachute Attack, and Stationary did meet parametric assumptions. There was a significant difference between white versus red light for Stationary behavior (White, 28.6%; Red, 70.7%; Wilcoxon, $N = 9$, $W = 44$, $P = 0.011$) and Unknown behavior (White, 7.67%; Red, 1.26%; Wilcoxon, $N = 9$, $W = 8.5$, $P = 0.047$). Arm Exploration median (IQR) in white light was (0.039 (0.062)) and median (IQR) in red light was (0.057 (0.108)) (T-test, $N = 9$, $t = 1.05$, $df = 8.41$, $P = 0.325$), Crawling median (IQR) in white light was (0.400 (0.289)) and median (IQR) in red light was (0.349 (0.255)) (T-test, $N = 9$, $t = -1.38$, $df = 11.9$, $P = 0.194$), and Unknown median (IQR) in white light was (0.067 (0.177)) and median (IQR) in red light was (0.013 (0.011)) (T-test, $N = 9$, $t = -2.29$, $df = 6.19$, $P = 0.061$) deeming the behaviors non-parametric. Bipedal Locomotion median (IQR) in white light was (0.013 (0.072)) and median (IQR) in red light was (0.0051 (0.019)) (Wilcoxon, $N = 9$, $W = 19.5$, $P = 0.549$), Burrowing median (IQR) in white light was (0.000 (0.000)) and median (IQR) in red light was (0.000 (0.000)) (Wilcoxon, $N = 9$, $W = 21$, $P = 0.391$), Entering Hole median (IQR) in white light was (0.061 (0.071)) and median (IQR) in red light was (0.017 (0.071)) (Wilcoxon, $N = 9$, $W = 16.5$, $P = 0.333$), Jetting median (IQR) in white light was (0.0476 (0.004)) and median (IQR) in red light was (0.004 (0.072)) (Wilcoxon, $N = 9$, $W = 17$, $P = 0.360$), Parachute Attack median (IQR) in white light was (0.000 (0.000)) and median (IQR) in red light was (0.000 (0.000)) (Wilcoxon, $N = 9$, $W = 28$, $P = 0.391$), and Stationary median (IQR) in white light was (0.275 (0.133)) and median (IQR) in red light was (0.776 (0.539)) (Wilcoxon, $N = 9$, $W = 44$, $P = 0.011$) classifying the behaviors as meeting parametric assumptions (Figure 6).

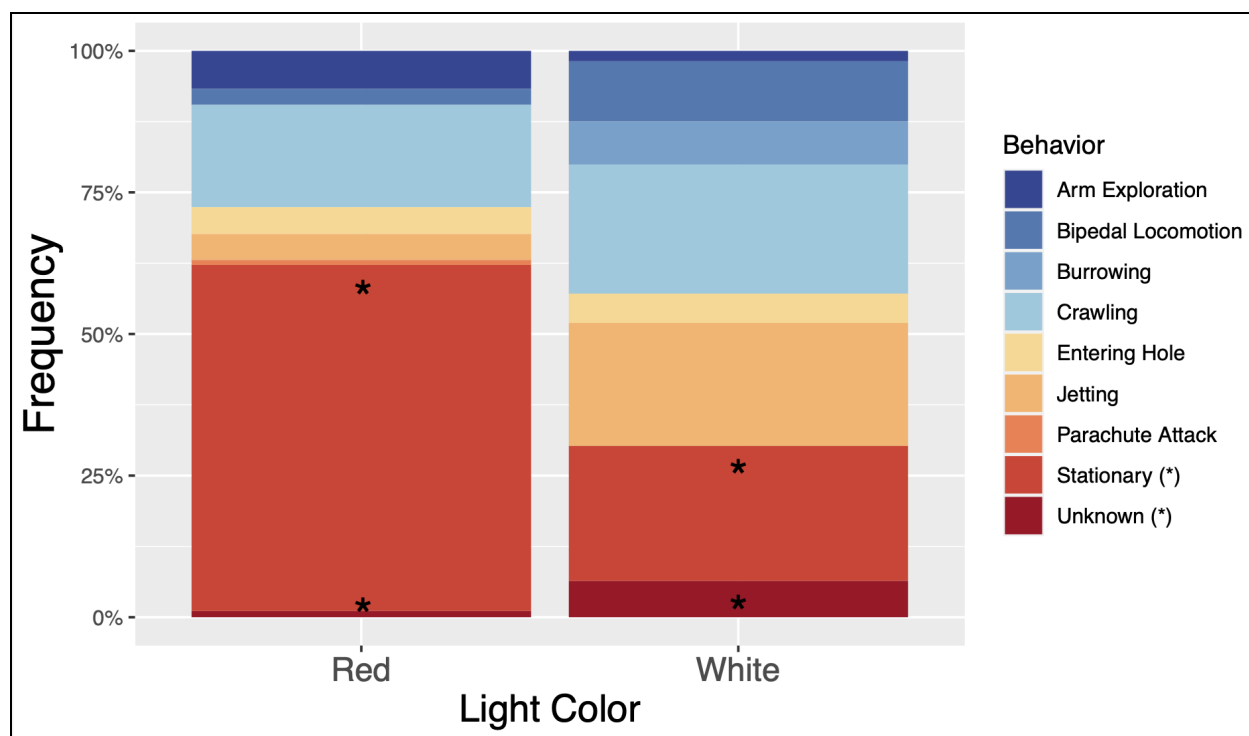


Figure 6. Stacked bar plot showing the relative percentages of behavior based on light color (White and Red). Each color represents a different behavior (Arm Exploration, Bipedal Locomotion, Burrowing, Crawling, Entering Hole, Jetting, Parachute Attack, Stationary, and Unknown). Asterisks (*) represent counts significantly higher than the expected counts ($P < 0.050$). No asterisks indicate no significantly higher difference between observed and expected counts.

4. Discussion

We examined whether *C. furvus* skin patterning was related to their behavior and substrate, and how *C. furvus* behavior was altered by white light compared to red light. We hypothesized that specific skin patterns would be associated with specific behaviors and substrate types, and that *C. furvus* will present more defense-driven behaviors under white light compared to red light. These hypotheses were partially supported by our results, with certain skin patterns presenting more often than expected with certain behaviors and substrates, and red light resulting in more natural behaviors.

4.1 The Effect of Substrate on Skin Pattern

Our findings partially support our hypothesis that *C. furvus* will change its skin pattern to lighter colors to match lighter-colored substrates and red to blend in with more colorful substrates in order to better camouflage with its surroundings. On substrates like Bedrock or Boulder, Green Algae, and Red Algae, which are typically a mixture of lightly colored and dark substrates, *C. furvus* was observed to have lighter patterns like Mottle and Stippled to help it better blend in with its mixed-colored environment. Also, it appeared that *C. furvus* will try to blend in with red and brown algae by imitating its surrounding substrates using skin patterns like Brick Red and White Elements and Brick Red with White Arm Spots and its many arms to mimic the algae. However, on substrates like Sand and Sparse Seagrass, *C. furvus* also appeared to have skin patterns like Brick Red and White Elements and Brick Red with White Arm Spots more often. This was interesting because the brick red color pattern made the octopus stand out against the background, contrasting with another co-occurring nocturnal octopus, *Octopus briareus*, which uses its uniform light blue-green skin pattern over light Sand patches (Hanlon and Wolterding, 1989). We hypothesize that *C. furvus* appears more brick red in lighter substrates to seem more intimidating to predators. As nocturnal species, the vivid brick red patterning appears as a bold and dark outline, mimicking potentially threatening species and objects (i.e. corals, rocks and boulders), serving as a warning signal for predators to stay away. Being able to change skin patterns to mimic or camouflage with different substrates helps minimize the risk of predation and can be used as an

anti-predator defense mechanism while performing other tasks, such as hunting, foraging and changing habitat locations (Hanlon and Messenger, 2018). Our results suggest that *C. furvus*' camouflage tactics vary greatly across substrates, making it difficult to conclude the implications of our findings without understanding its behavior and reasons for abrupt skin pattern changes. For instance, there were moments when *C. furvus* would try to blend in with lighter-colored substrates, mimic and camouflage with the red and brown algae, or completely stand out and appear intimidating. This demonstrates that *C. furvus* make moment-to-moment decisions about their skin patterning and the best camouflage tactics needed as they move through the environment. In the future, it is important to recognize the skin patterns octopus already display but acknowledge that they are constantly co-evolving with their environment in order to avoid predation and strengthen survival chances.

4.2 The Relationship between Behavior and Skin Pattern

The results partially supported our hypothesis that *C. furvus* would utilize lighter skin patterns when attempting to avoid detection by other organisms, and appear redder when foraging or escaping as an intimidation tactic. Burrowing and Parachute Attack were only observed in one individual, so these behaviors cannot be analyzed until a larger sample size is gathered. The skin pattern Brick Red and White Elements was displayed more than expected for Bipedal Locomotion and Entering Holes, partially supporting the theory that *C. furvus* will appear lighter when attempting to hide, presenting as multi-colored. This likely helps them perform crypsis and avoid detection by other organisms as they blend in with lighter substrates during the day and the dark waters at night. Furthermore, the skin pattern Mottle was present more often than expected while Stationary, further supporting the possibility of altered skin pattern as a form of crypsis, as they are able to appear as part of a larger variety of backgrounds or objects while in a multi-colored form. These lighter skin patterns may also act as deimatic behaviors, where octopuses mimic more threatening species, potentially intimidating and scaring away other predators (Hanlon and Messenger, 2018). This is similar to another species of octopus that occurs in the region, *Octopus insularis*, which also displays a Mottle pattern during Arm Exploration and Crawling

(Leite et al., 2009). This suggests that Mottle patterns could have higher prevalence during these exploratory behaviors in Western Atlantic octopus species. Mottle may help octopuses better prevent detection as the wider range of colors between red and white helps them blend in with a larger variety of substrates and mimic a wider number of objects. Because of the potential for higher Mottle frequencies, this skin pattern could also act as an important communication form, possibly to scare off predators through intimidation as part of a deimatic display, where less dangerous animals act as more dangerous ones, for example changing skin color or texture to appear as another poisonous creature (Hanlon and Messenger, 2018). When Jetting, *C. furvus* displayed Polka Dots and Brick Red and White Elements most often, supporting the hypothesis that they will have more vivid coloration when escaping, likely as a deimatic behavior, which could intimidate the predator (Hanlon and Messenger, 2018). Further steps should be taken to develop technological tools, like artificial intelligence softwares, to more accurately recognize these behaviors and patterns under red light conditions. Because of the increased identification difficulties associated with red light, it is important to explore new methods while continuing to ensure minimal disturbance to octopuses.

4.3 The Effect of White versus Red Light on Behavior

It was determined that Stationary proved to be the most prevalent behavior under red light compared to white light. These results supported our hypothesis that *C. furvus* will display more typical behaviors such as Arm Exploration, Crawling, or Stationary, when filmed with red light as opposed to white light. When *C. furvus* engages in stop-and-go hunting, Stationary is the main behavior exhibited by the species. As Stationary behavior occurs most often under red light, this supports the idea that red light is less disruptive to octopuses, allowing them to continue exhibiting natural behaviors without disturbance. However, because no other behavior category, especially Arm Exploration and Crawling, differed it is still possible that observation under red light is disruptive to octopuses. In the future, researchers should use red light but also implement techniques to make observers less conspicuous to octopuses, ultimately increasing the chances of capturing the truest form of natural octopus behavior.

4.4 Limitations

A few methodological factors could have impacted our results and should be addressed in future studies. White light was utilized while searching for *C. furvus*, only switching to red light once an octopus was found. The time to switch from white to red light may have been long enough that the octopus was disturbed, affecting its behavior. Additionally, while diving, researchers noticed that octopuses appeared to be reacting to the loud noise that tank bangers made. Other methods of alerting divers should be tested to determine what methods cause the fewest defense-driven reactions of octopus.

4.5 Conclusions

Several of the six skin patterns in *C. furvus* were associated with certain behaviors and substrate types, suggesting that *C. furvus*' skin patterning is influenced by both factors. Additionally, Stationary behavior was determined to be the prevailing behavior while filming with red light, likely reflecting more natural foraging behavior and supporting the idea that the use of red light is better for more accurate observations of this species. Because octopuses act as important middle-level predators and as a bridge between trophic levels, increased behavioral knowledge is crucial to maintaining a well-balanced ecosystem. Greater knowledge of skin patterning, behaviors and their correlations to movement, aids in interpreting interactions with other organisms and promoting future behavioral research of octopuses.

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